

CS305 Database Systems
Lecture 18: SQL and Normalization
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Outline

1. Relational model
2. Keys and joins
3. Normalization
4. Transactions

The Relational Model

- A relation is a table of rows and columns.
- Each row is a tuple representing one entity.
- Each column is an attribute with a domain.

Primary Key

- A primary key uniquely identifies each row in a table.
- It cannot contain NULL values.

Foreign Key

- A foreign key references the primary key of another table.
- It enforces referential integrity between relations.

The SELECT Statement

- 1. SELECT retrieves rows from one or more tables.
- 1. The WHERE clause filters rows by a predicate.
- 1. SELECT name FROM users WHERE id = 1;

Projection

2) Projection chooses which columns appear in the result.

2) It is written as the column list after SELECT.

Aggregate Functions

- * An aggregate function computes a single value over many rows.
- * COUNT returns the number of rows in a group.

Grouping Rows

- GROUP BY partitions rows into groups sharing a value.
- Aggregates are then computed per group.

The HAVING Clause

(3) HAVING filters groups after aggregation.

(3) WHERE filters rows before grouping.

Inner Join

- An inner join returns rows matching in both tables.
- The join condition compares a foreign key to a primary key.

Outer Join

- A left outer join keeps unmatched rows from the left table.
- Missing values are padded with NULL.

Subqueries

- * A subquery is a query nested inside another query.
- * A correlated subquery references the outer query.

Views

1. A view is a named query stored in the schema.
1. It presents derived data as if it were a table.



Functional Dependency

2) A functional dependency means one attribute determines another.

2) It is written $X \rightarrow Y$.

First Normal Form

- First normal form requires every attribute to be atomic.
- Repeating groups are not allowed.

Second Normal Form

- Second normal form removes partial dependencies on a composite key.
- Every non-key attribute depends on the whole key.

Third Normal Form

- * Third normal form removes transitive dependencies.
- * Non-key attributes must not determine other non-key attributes.

Denormalization

- Denormalization reintroduces redundancy to improve read speed.
- It is a deliberate trade-off against update anomalies.

Transactions

1. A transaction is a unit of work executed atomically.
1. It either commits entirely or aborts entirely.

Atomicity

- Atomicity means all operations succeed or none do.
- A partial update is never visible.

Durability

- Durability means committed changes survive failure.
- The log is flushed before commit returns.

Isolation Levels

- * Isolation determines how concurrent transactions see each other.
- * Serializable is the strictest level.

Deadlock

- A deadlock occurs when transactions wait on each other forever.
- The system resolves it by aborting one transaction.

Stored Procedures

- 2) A stored procedure is program logic stored in the database.
- 2) It reduces round trips between client and server.

Comparison Table

Isolation Level	Dirty Read	Lost Update
Read Committed	No	Yes
Serializable	No	No

- Serializable forbids every anomaly listed above.

Worked Example

```
SELECT dept, COUNT(*) FROM staff GROUP BY dept HAVING COUNT(*) > 5;
```

- The HAVING clause removes small departments from the result.

Summary

- Keys, joins, normalization and transactions were covered.



Appendix

- Additional reading is listed here.

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